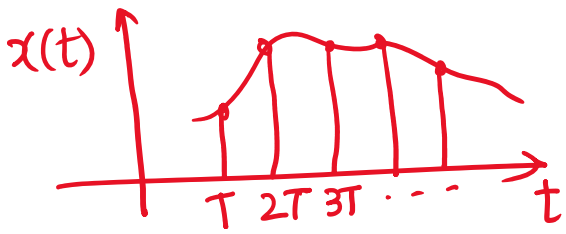
Sampling:

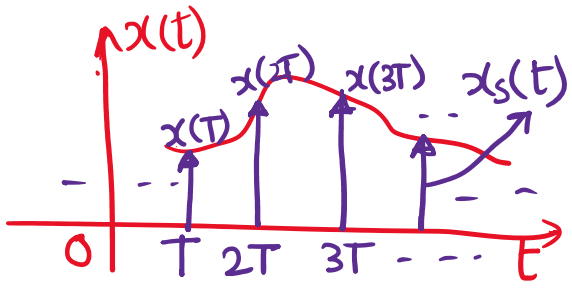
$x(t) \longrightarrow x[n]$

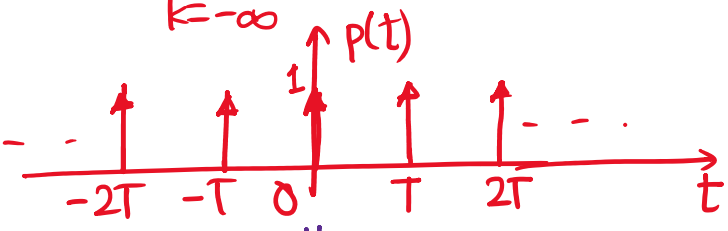


Nyquist Sampling Theorem: (Ideal Impulse Sampling)

Let $x(t)$ be sampled using $p(t) = \sum_{k=-\infty}^{\infty} \delta(t-kT)$

s.t. $x_s(t) = x(t)p(t)$





$\Downarrow$

$$p(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad a_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t}\, dt$$

$$\therefore p(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t} \quad = \frac{1}{T} \ \forall k$$

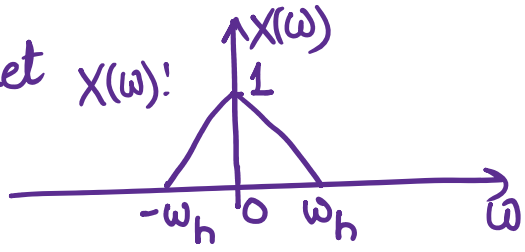
$$\Rightarrow P(\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} 2\pi\, \delta(\omega - k\omega_0)$$

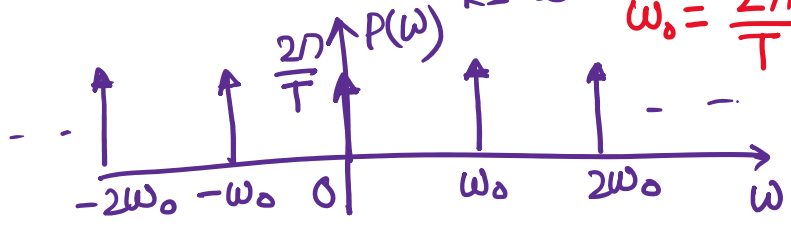
$$\omega_0 = \frac{2\pi}{T}$$

$$x_s(t) = x(t)\,p(t)$$

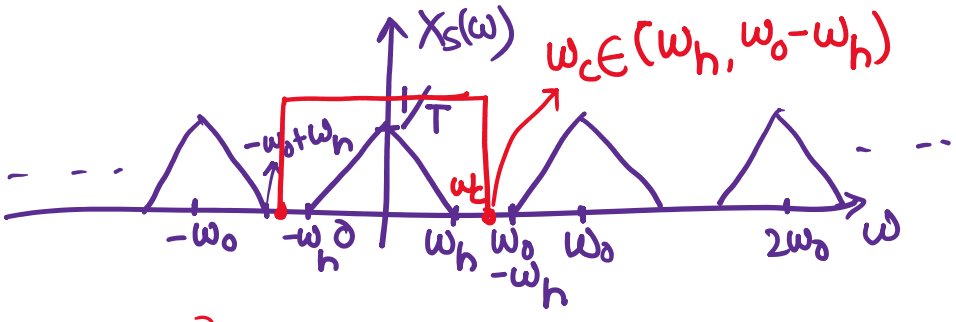
$$\Rightarrow X_s(\omega) = \frac{1}{2\pi} X(\omega) * P(\omega)$$

Let $X(\omega)$:





$\therefore X_s(\omega)$:



$\omega_h < \omega_0 - \omega_h$ } Non-overlap

$\therefore \omega_0 > 2\omega_h$

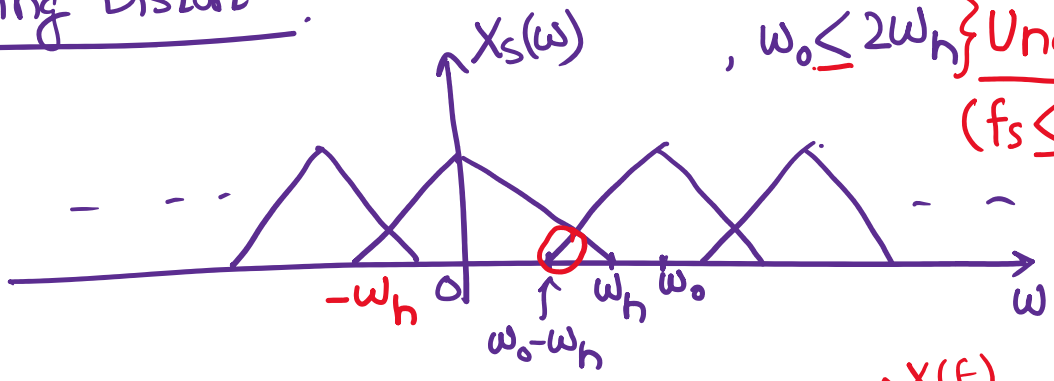
$\omega_h < \omega_0 - \omega_h$ } Non-overlap

$\Rightarrow \omega_0 > 2\omega_h$

Nyquist Sampling Theorem :

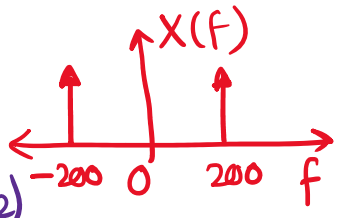
For a signal bandlimited to f_h, the signal can be reconstructed perfectly from its sampled version, provided sampling freq. $f_s > 2f_h$

Aliasing Distortⁿ :

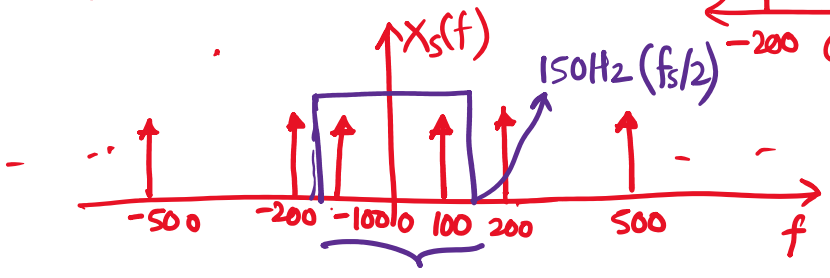


, $\omega_0 \leq 2\omega_h$ } Undersampling ($f_s \leq 2f_H$)

Ex. $x(t) = \cos(400\pi t)$, $f_s = 300\,Hz$



$X_s(f)$



$\therefore$ 200 Hz $\xrightarrow{\text{Alias}}$ 100 Hz } Aliasing

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Wheel Rotⁿ at (a) 30π rad/s, (b) 90π rad/s (c) ? (Clockwise rotⁿ)
↓ stationary

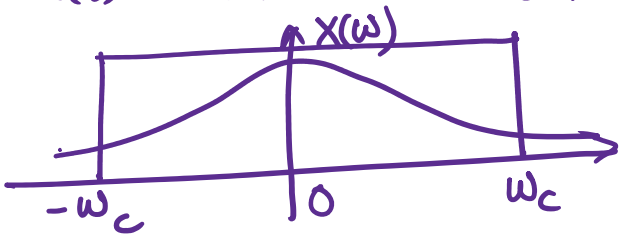
& f_s : (a) 60 fps & 30π rad/s
⇓

Limitatⁿ of Ideal Impulse Sampling :

1) Ideal Impulse are not realizable.
✓2) " LPF " " .

✓2) " LPF " " .

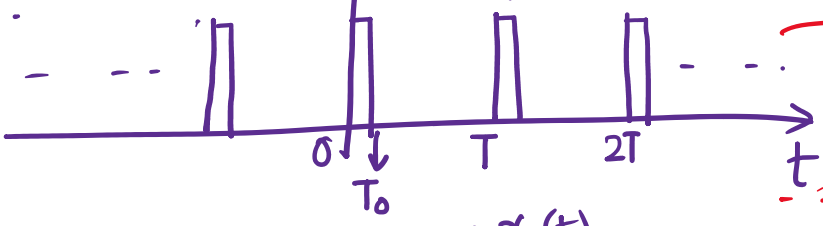
✓3) $x(t)$ can't be bandlimited (as $x(t)$ is always timelimited in real life)

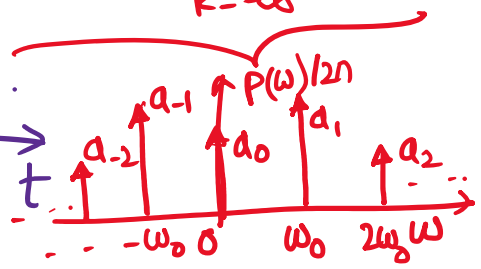


2) Natural Sampling:

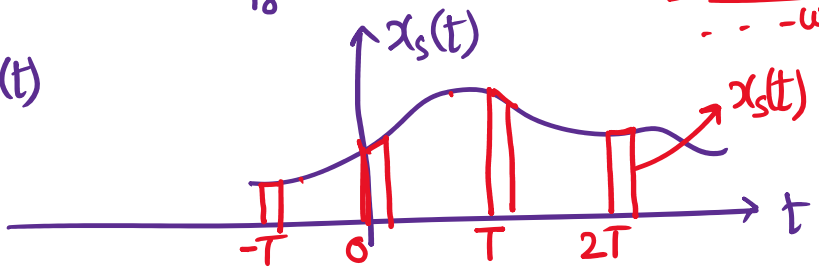
$$p(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \Rightarrow P(\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \, \delta(\omega - k\omega_0)$$

$p(t)$:

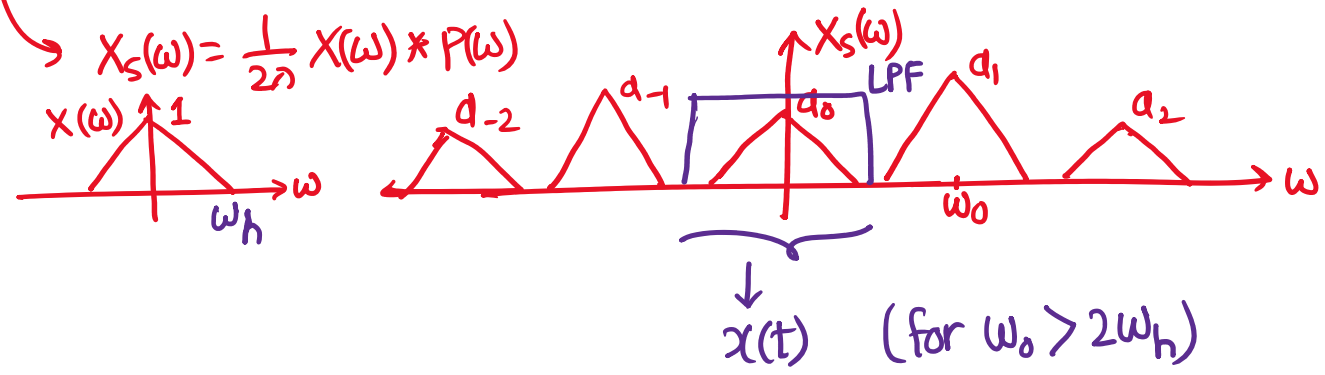




$x_s(t) = x(t)p(t)$



$$X_s(\omega) = \frac{1}{2\pi} X(\omega) * P(\omega)$$



(for $\omega_0 > 2\omega_h$)

Any periodic $p(t)$ works, if $a_0 \neq 0$